6.Write a case study to identify and analyze the requirements of Food Delivery Applications, and prepare positive test scenarios as well as test cases for the application.

Aim

To identify and analyze the requirements of a Food Delivery application and prepare positive test scenarios and test cases.

Requirements Analysis

Functional requirements

1. Customer registration and login.
2. Restaurant and food search.
3. Viewing menus, prices, and availability.
4. Adding food items to the cart.
5. Applying valid discount coupons.
6. Selecting a delivery address.
7. Placing orders and making payments.
8. Tracking orders and viewing order history.

Non-functional requirements

- Secure user and payment information.
- Accurate pricing and order calculations.
- Reliable order processing.
- User-friendly interface and acceptable response time.

Positive Test Scenarios

Scenario ID	**Test scenario**	**Expected result**
FDS01	Login with valid credentials	Login successful
FDS02	Search for an available restaurant	Matching restaurants displayed
FDS03	View a restaurant menu	Menu and prices displayed correctly

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FDS04	Add available food to cart	Food item added successfully
FDS05	Increase food quantity	Cart total updated correctly
FDS06	Apply a valid coupon	Eligible discount applied
FDS07	Place an order with a valid address	Order placed successfully
FDS08	Complete payment successfully	Payment confirmed
FDS09	Track an existing order	Correct order status displayed
FDS10	View order history	Previous orders displayed

Positive Test Cases

Test Case ID: FD_TC01

1. Test case: Login with valid credentials.
2. Precondition: Customer has a registered account.
3. Steps to execute:
    1. Open the application.
    2. Enter the registered email and password.
    3. Click Login.
4. Test data: Valid email and password.
5. Expected result: Customer logs in successfully.
6. Status: Pass/Fail

Test Case ID: FD_TC02

1. Test case: Search for an available restaurant.
2. Precondition: Restaurants are available in the selected delivery location.
3. Steps to execute:
    1. Open the application.
    2. Enter a restaurant name in the search bar.
    3. Click Search.

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4. Test data: Existing restaurant name.
5. Expected result: Matching restaurant details are displayed.
6. Status: Pass/Fail

Test Case ID: FD_TC03

1. Test case: View restaurant menu.
2. Precondition: Restaurant is open and its menu is available.
3. Steps to execute:
    1. Select a restaurant.
    2. Open its menu.
    3. Select a food category.
4. Test data: Available restaurant and menu.
5. Expected result: Correct food names, prices, and availability are displayed.
6. Status: Pass/Fail

Test Case ID: FD_TC04

1. Test case: Add food items to the cart.
2. Precondition: A food item is available.
3. Steps to execute:
    1. Select a food item.
    2. Click Add to Cart.
    3. Open the cart.
4. Test data: One available food item.
5. Expected result: The selected item appears in the cart with the correct price.
6. Status: Pass/Fail

Test Case ID: FD_TC05

1. Test case: Update food quantity.

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2. Precondition: Cart contains a food item.
3. Steps to execute:
    1. Open the cart.
    2. Increase quantity from 1 to 2.
    3. Check the subtotal.
4. Test data: Food item priced at ₹150, quantity 2.
5. Expected result: Subtotal is ₹300, excluding additional charges and discounts.
6. Status: Pass/Fail

Test Case ID: FD_TC06

1. Test case: Apply a valid discount coupon.
2. Precondition: Cart meets the coupon's eligibility conditions.
3. Steps to execute:
    1. Open the cart.
    2. Enter a valid coupon.
    3. Click Apply.
4. Test data: Valid, eligible coupon.
5. Expected result: The applicable discount is calculated correctly.
6. Status: Pass/Fail

Test Case ID: FD_TC07

1. Test case: Place an order with a valid delivery address.
2. Precondition: Cart contains available food items.
3. Steps to execute:
    1. Open the cart.
    2. Select Checkout.
    3. Enter a valid delivery address.

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        4. Confirm the order.
    4. Test data: Valid address and contact details.
    5. Expected result: Order is created with a unique order ID.
    6. Status: Pass/Fail

Test Case ID: FD_TC08

1. Test case: Complete payment successfully.
2. Precondition: Order is ready for payment.
3. Steps to execute:
    1. Select a payment method.
    2. Complete payment using the payment test environment.
    3. Check the order status.
4. Test data: Successful test payment response.
5. Expected result: Payment is confirmed and the order status is updated correctly.
6. Status: Pass/Fail

Test Case ID: FD_TC09

1. Test case: Track an order.
2. Precondition: An order exists.
3. Steps to execute:
    1. Open My Orders.
    2. Select the current order.
    3. Click Track Order.
4. Test data: Valid order ID.
5. Expected result: The correct preparation and delivery status is displayed.
6. Status: Pass/Fail

Test Case ID: FD_TC10

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1. Test case: View order history.
2. Precondition: Customer has previous orders.
3. Steps to execute:
    1. Log in.
    2. Open My Orders.
4. Test data: Registered customer account with order history.
5. Expected result: Previous orders and their details are displayed.
6. Status: Pass/Fail

Result

The requirements of the Food Delivery application were analyzed, and ten positive test scenarios and test cases were prepared to verify restaurant search, menu browsing, cart operations, ordering, payment, and delivery tracking.

7.Write a case study to identify and analyze the requirements of Food Delivery Applications, and prepare negative test scenarios as well as test cases for the application.

Aim

To identify and analyze the requirements of a Food Delivery application and prepare negative test scenarios and test cases.

Requirements Analysis

Functional requirements

1. Validate login and registration details.
2. Check restaurant and food availability.
3. Validate quantities and cart totals.
4. Verify coupon eligibility.
5. Require valid delivery details.
6. Handle failed payments.

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7. Prevent unauthorized order access.
8. Prevent duplicate orders.

Non-functional requirements

- Protect customer information.
- Display meaningful error messages.
- Avoid duplicate charges.
- Maintain accurate order and payment records.

Negative Test Scenarios

Scenario ID	Test scenario	Expected result
FD_NS01	Login with an incorrect password	Login rejected
FD_NS02	Register with an invalid email	Validation error displayed
FD_NS03	Search for a nonexistent restaurant	No matching restaurant displayed
FD_NS04	Add an unavailable food item to cart	Item cannot be ordered
FD_NS05	Enter zero or negative quantity	Invalid quantity rejected
FD_NS06	Apply an expired coupon	Coupon rejected
FD_NS07	Checkout without a delivery address	Checkout blocked
FD_NS08	Payment fails	Order is not marked as paid
FD_NS09	Order from a restaurant outside the delivery area	Order rejected
FD_NS10	Submit the same order repeatedly	Duplicate order prevented
FD_NS11	Access another customer's order	Access denied
FD_NS12	Checkout with an empty cart	Checkout blocked

Negative Test Cases

Test Case ID: FD_NTC01

1. Test case: Login with an incorrect password.
2. Precondition: Customer has a registered account.

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3. Steps to execute:
    1. Open the login page.
    2. Enter a registered email and incorrect password.
    3. Click Login.
4. Test data: Valid email and wrong password.
5. Expected result: Login is rejected and an error message is displayed.
6. Status: Pass/Fail

Test Case ID: FD_NTC02

1. Test case: Register with an invalid email.
2. Precondition: Registration page is available.
3. Steps to execute:
    1. Open registration.
    2. Enter an invalid email.
    3. Complete the remaining fields.
    4. Submit the form.
4. Test data: abc123
5. Expected result: The application rejects the invalid email.
6. Status: Pass/Fail

Test Case ID: FD_NTC03

1. Test case: Search for a nonexistent restaurant.
2. Precondition: Restaurant search is available.
3. Steps to execute:
    1. Open the application.
    2. Enter a nonexistent restaurant name.
    3. Click Search.

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4. Test data: XYZRestaurant123
5. Expected result: A no-results message is displayed without crashing.
6. Status: Pass/Fail

Test Case ID: FD_NTC04

1. Test case: Add an unavailable food item to the cart.
2. Precondition: Restaurant menu is available.
3. Steps to execute:
    1. Open the restaurant menu.
    2. Select an item marked unavailable.
    3. Attempt to add it to the cart.
4. Test data: Unavailable food item.
5. Expected result: The application prevents ordering the unavailable item.
6. Status: Pass/Fail

Test Case ID: FD_NTC05

1. Test case: Enter zero or negative food quantity.
2. Precondition: Cart contains a food item.
3. Steps to execute:
    1. Open the cart.
    2. Enter quantity 0 or -2.
    3. Click Update.
4. Test data: 0, -2
5. Expected result: Invalid quantities are rejected.
6. Status: Pass/Fail

Test Case ID: FD_NTC06

1. Test case: Apply an expired coupon.

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2. Precondition: Cart contains food items.
3. Steps to execute:
    1. Open the cart.
    2. Enter an expired coupon.
    3. Click Apply.
4. Test data: Expired coupon code.
5. Expected result: Coupon is rejected and no discount is applied.
6. Status: Pass/Fail

Test Case ID: FD_NTC07

1. Test case: Checkout without a delivery address.
2. Precondition: Cart contains an item.
3. Steps to execute:
    1. Open Checkout.
    2. Leave the delivery address empty.
    3. Click Continue.
4. Test data: Empty delivery address.
5. Expected result: Checkout is blocked and a validation message is displayed.
6. Status: Pass/Fail

Test Case ID: FD_NTC08

1. Test case: Handle payment failure.
2. Precondition: An order is ready for payment.
3. Steps to execute:
    1. Proceed to payment.
    2. Simulate a failed payment using the payment test environment.
    3. Check the payment and order status.

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4. Test data: Failed payment response.
5. Expected result: Payment is not marked successful, and the order is not marked as paid.
6. Status: Pass/Fail

Test Case ID: FD_NTC09

1. Test case: Order from outside the delivery area.
2. Precondition: Restaurant has a defined delivery area.
3. Steps to execute:
    1. Select a restaurant.
    2. Enter a delivery address outside its service area.
    3. Attempt checkout.
4. Test data: Address outside the supported delivery zone.
5. Expected result: The application rejects the order and displays an appropriate message.
6. Status: Pass/Fail

Test Case ID: FD_NTC10

1. Test case: Prevent duplicate order submission.
2. Precondition: Checkout is ready for order placement.
3. Steps to execute:
    1. Click Place Order.
    2. Repeat the same submission or retry after a timeout.
    3. Check order history.
4. Test data: Same checkout request submitted twice.
5. Expected result: Only one order is created for the same transaction.
6. Status: Pass/Fail

Test Case ID: FD_NTC11

1. Test case: Access another customer's order.

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2. Precondition: Customer is logged in.
3. Steps to execute:
    1. Log in with a customer account.
    2. Attempt to access an order belonging to another customer.
4. Test data: Unauthorized order ID.
5. Expected result: Access is denied and customer details remain private.
6. Status: Pass/Fail

Test Case ID: FD_NTC12

1. Test case: Checkout with an empty cart.
2. Precondition: Shopping cart is available.
3. Steps to execute:
    1. Remove all items from the cart.
    2. Click Checkout.
4. Test data: Empty cart.
5. Expected result: Checkout is blocked with an appropriate message.
6. Status: Pass/Fail

Result

Twelve negative test scenarios and test cases were prepared to validate input handling, food availability, coupon validation, delivery restrictions, payment failures, and order security.

8.Write a case study to identify and analyze the requirements of Banking Applications, and prepare positive test scenarios as well as test cases for the application.

Aim

To identify and analyze the requirements of a Banking application and prepare positive test scenarios and test cases.

Requirements Analysis

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Functional requirements

1. Customer login and authentication.
2. Account balance enquiry.
3. Fund transfer between accounts.
4. Viewing transaction history and statements.
5. Bill payment.
6. Beneficiary management.
7. Updating permitted profile details.
8. Secure logout.

Non-functional requirements

- Secure authentication and transaction processing.
- Accurate account balances.
- Reliable transaction records.
- Acceptable response time and availability.

Positive Test Scenarios

Scenario ID	**Test scenario**	**Expected result**
BA_PS01	Login with valid credentials	Login successful
BA_PS02	Check account balance	Correct balance displayed
BA_PS03	Transfer funds to a valid beneficiary	Transfer successful
BA_PS04	View transaction history	Correct transactions displayed
BA_PS05	Download bank statement	Statement downloaded
BA_PS06	Pay a bill using valid details	Payment successful
BA_PS07	Add a valid beneficiary	Beneficiary added
BA_PS08	Update permitted profile details	Details updated

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BA_PS09	Log out of the application	Session terminated

Positive Test Cases

Test Case ID: BA_TC01

1. Test case: Login with valid credentials.
2. Precondition: Customer has an active online banking account.
3. Steps to execute:
    1. Open the banking application.
    2. Enter the registered username and password.
    3. Complete any required authentication.
    4. Click Login.
4. Test data: Valid login credentials.
5. Expected result: The customer dashboard is displayed.
6. Status: Pass/Fail

Test Case ID: BA_TC02

1. Test case: Check account balance.
2. Precondition: Customer is logged in.
3. Steps to execute:
    1. Open Accounts.
    2. Select the required account.
    3. Click Balance Enquiry.
4. Test data: Valid account.
5. Expected result: The correct account balance is displayed.
6. Status: Pass/Fail

Test Case ID: BA_TC03

1. Test case: Transfer funds to a valid beneficiary.

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2. Precondition: Customer is authenticated, the beneficiary is valid, and sufficient funds are available.
3. Steps to execute:
    1. Open Fund Transfer.
    2. Select the source account.
    3. Select a registered beneficiary.
    4. Enter the transfer amount.
    5. Confirm the transaction using the required authentication.
4. Test data: Valid beneficiary and ₹1,000 transfer amount, with sufficient funds.
5. Expected result: The transfer is completed, the account balance is updated, and a transaction reference is generated.
6. Status: Pass/Fail

Test Case ID: BA_TC04

1. Test case: View transaction history.
2. Precondition: Customer has transaction records.
3. Steps to execute:
    1. Open Transaction History.
    2. Select a date range.
    3. Click Search.
4. Test data: Valid date range.
5. Expected result: Correct transactions within the selected period are displayed.
6. Status: Pass/Fail

Test Case ID: BA_TC05

1. Test case: Download bank statement.
2. Precondition: Customer is logged in and statement data is available.

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3. Steps to execute:
    1. Open Statements.
    2. Select an account and date range.
    3. Click Download.
4. Test data: Valid account and date range.
5. Expected result: The statement downloads successfully with correct transaction details.
6. Status: Pass/Fail

Test Case ID: BA_TC06

1. Test case: Pay a bill successfully.
2. Precondition: Customer has sufficient funds and a valid biller is available.
3. Steps to execute:
    1. Open Bill Payments.
    2. Select a biller.
    3. Enter valid bill details.
    4. Confirm payment.
4. Test data: Valid bill reference and amount.
5. Expected result: Payment is processed, the debit is recorded, and a confirmation reference is displayed.
6. Status: Pass/Fail

Test Case ID: BA_TC07

1. Test case: Add a valid beneficiary.
2. Precondition: Customer is authenticated.
3. Steps to execute:
    1. Open Beneficiary Management.
    2. Enter valid beneficiary details.

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3. Complete any required verification.
4. Click Save.

4. Test data: Valid beneficiary information.
5. Expected result: Beneficiary is added according to the bank's verification rules.
6. Status: Pass/Fail

Test Case ID: BA_TC08

1. Test case: Update permitted profile details.
2. Precondition: Customer is logged in.
3. Steps to execute:
    1. Open Profile Settings.
    2. Update an editable field.
    3. Save the changes.
4. Test data: Valid updated contact details.
5. Expected result: Changes are saved and displayed correctly.
6. Status: Pass/Fail

Test Case ID: BA_TC09

1. Test case: Logout securely.
2. Precondition: Customer is logged in.
3. Steps to execute:
    1. Click Logout.
    2. Attempt to reopen a protected page.
4. Test data: Active customer session.
5. Expected result: The session ends and authentication is required to access protected pages.
6. Status: Pass/Fail

Result

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The banking application's requirements were analyzed, and nine positive test scenarios and test cases were prepared to verify authentication, balance enquiry, fund transfer, statements, bill payment, beneficiary management, and logout.

9.Write a case study to identify and analyze the requirements of Banking Applications, and prepare negative test scenarios as well as test cases for the application.

Aim

To identify and analyze the requirements of a Banking application and prepare negative test scenarios and test cases.

Requirements Analysis

Functional requirements

1. Validate customer login.
2. Reject invalid transaction details.
3. Check available funds before transfers.
4. Validate beneficiary information.
5. Handle failed payments and interrupted transactions.
6. Restrict access to unauthorized accounts.
7. Protect transactions against duplicate processing.
8. Validate bill payments and profile updates.

Non-functional requirements

- Strong authentication and access control.
- Accurate and consistent financial records.
- Secure handling of sensitive information.
- Reliable recovery from transaction failures.

Negative Test Scenarios

Scenario ID	Test scenario	Expected result

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BA_NS01	Login with an incorrect password	Login rejected
BA_NS02	Submit login with empty fields	Validation displayed
BA_NS03	Transfer more money than the available balance	Transfer rejected
BA_NS04	Enter a zero or negative transfer amount	Amount rejected
BA_NS05	Enter an invalid beneficiary account number	Transfer rejected
BA_NS06	Submit a transfer without required authentication	Transfer blocked
BA_NS07	Access another customer's account	Access denied
BA_NS08	Pay a bill with invalid bill details	Payment rejected
BA_NS09	Repeat the same transfer request	Duplicate transfer prevented
BA_NS10	Interrupt a transfer during processing	Transaction remains consistent
BA_NS11	Access banking pages after logout	Authentication required

Negative Test Cases

Test Case ID: BA_NTC01

1. Test case: Login with an incorrect password.
2. Precondition: Customer has a registered account.
3. Steps to execute:
    1. Open the banking application.
    2. Enter a valid username and incorrect password.
    3. Click Login.
4. Test data: Valid username and incorrect password.
5. Expected result: Login is rejected and an appropriate error message is displayed.
6. Status: Pass/Fail

Test Case ID: BA_NTC02

1. Test case: Submit login with empty fields.
2. Precondition: Login page is displayed.

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3. Steps to execute:
    1. Leave the username and password fields empty.
    2. Click Login.
4. Test data: Empty username and password.
5. Expected result: Required-field validation appears and login is denied.
6. Status: Pass/Fail

Test Case ID: BA_NTC03

1. Test case: Transfer more money than the available balance.
2. Precondition: Customer is logged in and has a known available balance.
3. Steps to execute:
    1. Open Fund Transfer.
    2. Select a valid beneficiary.
    3. Enter an amount greater than the available balance.
    4. Submit the transfer.
4. Test data: Available balance ₹2,000; transfer amount ₹5,000.
5. Expected result: Transfer is rejected and no amount is debited.
6. Status: Pass/Fail

Test Case ID: BA_NTC04

1. Test case: Enter a zero or negative transfer amount.
2. Precondition: Fund Transfer page is available.
3. Steps to execute:
    1. Select a valid beneficiary.
    2. Enter 0 or -500 as the transfer amount.
    3. Submit the request.
4. Test data: 0, -500

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5. Expected result: Invalid transfer amount is rejected.
6. Status: Pass/Fail

Test Case ID: BA_NTC05

1. Test case: Enter an invalid beneficiary account number.
2. Precondition: Customer is logged in.
3. Steps to execute:
    1. Open Fund Transfer.
    2. Enter an invalid beneficiary account number.
    3. Enter an amount and submit.
4. Test data: Invalid or nonexistent beneficiary account.
5. Expected result: The system rejects the beneficiary details and does not process the transfer.
6. Status: Pass/Fail

Test Case ID: BA_NTC06

1. Test case: Submit a transfer without required authentication.
2. Precondition: A transfer requires transaction authentication.
3. Steps to execute:
    1. Enter valid transfer details.
    2. Skip or provide invalid transaction authentication.
    3. Attempt to confirm the transfer.
4. Test data: Missing or invalid required authentication.
5. Expected result: The transfer is blocked.
6. Status: Pass/Fail

Test Case ID: BA_NTC07

1. Test case: Access another customer's account.
2. Precondition: Customer is logged in to their own account.

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3. Steps to execute:
    1. Attempt to open another customer's account through an unauthorized request.
    2. Submit the request.
4. Test data: Unauthorized account identifier.
5. Expected result: Access is denied and confidential information is not displayed.
6. Status: Pass/Fail

Test Case ID: BA_NTC08

1. Test case: Pay a bill using invalid details.
2. Precondition: Customer is logged in.
3. Steps to execute:
    1. Open Bill Payments.
    2. Enter an invalid bill reference.
    3. Submit the payment.
4. Test data: Invalid bill reference.
5. Expected result: The payment is rejected and the account is not incorrectly debited.
6. Status: Pass/Fail

Test Case ID: BA_NTC09

1. Test case: Prevent duplicate fund transfers.
2. Precondition: A transfer request is ready to submit.
3. Steps to execute:
    1. Submit a transfer request.
    2. Repeat the same request or retry after a simulated timeout.
    3. Check transaction history and account balances.
4. Test data: Same transfer request submitted twice.
5. Expected result: Only one transfer is processed for the same transaction.

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6. Status: Pass/Fail

Test Case ID: BA_NTC10

1. Test case: Handle a network interruption during a transfer.
2. Precondition: A transfer can be tested in a controlled test environment.
3. Steps to execute:
    1. Initiate a valid transfer.
    2. Simulate a network interruption during processing.
    3. Reconnect and check transaction history and account balances.
4. Test data: Valid transfer and simulated network failure.
5. Expected result: The system must reconcile the transaction correctly, without an incorrect debit or duplicate transfer.
6. Status: Pass/Fail

Test Case ID: BA_NTC11

1. Test case: Access protected banking pages after logout.
2. Precondition: Customer is logged in.
3. Steps to execute:
    1. Click Logout.
    2. Reopen a protected page.
4. Test data: Ended session.
5. Expected result: The system requires authentication and does not reveal account information.
6. Status: Pass/Fail

Result

Eleven negative test scenarios and test cases were prepared to validate login security, transfer validation, beneficiary checks, payment handling, duplicate transaction prevention, and access control.

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10.Develop a comprehensive Test Plan document for a chosen application (such as a Library Management System), outlining various testing strategies, methodologies, and criteria to ensure thorough evaluation and validation of the system's functionality.

Aim

To develop a comprehensive Test Plan document for a Library Management System, covering testing strategies, methodologies, scope, resources, and completion criteria.

1. Introduction

A Library Management System is a software application used to manage books, members, book issuing, returns, renewals, reservations, and fines. The test plan defines how the system will be tested to verify that it meets its functional and non-functional requirements.

2. Objectives

- Verify that all library functions operate correctly.
- Identify and document software defects.
- Validate book availability, issue, and return operations.
- Ensure accurate fine calculation.
- Verify authentication and access control.
- Check usability, performance, and reliability.

3. Test Scope

In scope

1. User login and logout.
2. Book registration and catalogue search.
3. Member registration and management.
4. Book issue and return.
5. Book renewal and reservation.
6. Fine calculation and payment recording.
7. Report generation.

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8. Input validation and access control.

Out of scope

1. Testing external payment-provider infrastructure.
2. Testing physical barcode scanner hardware.
3. Testing unrelated third-party systems.

4. Testing Strategies and Methodologies

Testing type	**Purpose**
Functional testing	Verify that library functions meet requirements
Positive testing	Verify valid operations
Negative testing	Verify rejection of invalid inputs
Integration testing	Check interactions between member, book, and transaction modules
System testing	Test the complete application
Regression testing	Ensure fixes do not break existing features
Performance testing	Evaluate response time under expected load
Security testing	Verify authentication and access permissions
Usability testing	Check whether users can navigate the system easily
User acceptance testing	Confirm that the system meets library staff and member needs

5. Test Environment

- Hardware: Computer or laptop with network access.
- Software: Web browser, Library Management System, test database.
- Database: Sample book, member, and transaction records.
- Test tools: Manual test-case checklist, spreadsheet for defect tracking, and an automation tool if available.

6. Test Design Techniques

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1. Equivalence partitioning: Divide inputs into valid and invalid groups, such as valid and invalid member IDs.
2. Boundary value analysis: Test values at and around limits, such as maximum book-issue limits.
3. Decision table testing: Verify combinations of membership status, book availability, and issue limits.
4. State transition testing: Verify transitions between available, issued, returned, and reserved book states.
5. Error guessing: Test likely failure conditions, such as missing member details or duplicate book IDs.

7. Sample Test Cases

Test case ID	**Test scenario**	**Expected result**
LMS_TC01	Login with valid credentials	Dashboard displayed
LMS_TC02	Search for an existing book	Correct book details displayed
LMS_TC03	Issue an available book to an eligible member	Book issued successfully
LMS_TC04	Return an issued book	Book marked as available, subject to reservations
LMS_TC05	Calculate an overdue fine	Fine calculated according to library rules
LMS_TC06	Issue a book to an invalid member	Issue rejected
LMS_TC07	Issue an already-issued unavailable book	Issue rejected
LMS_TC08	Enter an invalid book ID	Error message displayed
LMS_TC09	Access administrator functions as a member	Access denied
LMS_TC10	Submit a book issue request without a member ID	Request rejected

8. Test Execution Strategy

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1. Review the system requirements.
2. Prepare test scenarios and test cases.
3. Set up the test environment and test data.
4. Execute positive and negative tests.
5. Record actual results and compare them with expected results.
6. Log defects with unique IDs, reproduction steps, severity, and priority.
7. Retest corrected defects.
8. Perform regression testing.
9. Prepare the final test summary report.

9. Entry Criteria

- Requirements are available and reviewed.
- The application is deployed in the test environment.
- Test data and test cases are prepared.
- Required test accounts and permissions are configured.

10. Exit Criteria

- All planned critical test cases have been executed.
- All critical and high-severity defects are resolved or formally accepted.
- Fixed defects have been retested.
- Regression testing is completed.
- Test results and unresolved issues are documented.

11. Deliverables

1. Test Plan document.
2. Test scenarios and test cases.
3. Test execution report.
4. Defect or bug reports.

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5. Retesting and regression results.
6. Final Test Summary Report.

12. Roles and Responsibilities

Role	**Responsibility**
Test Lead	Plan testing and review results
Tester	Design and execute test cases
Developer	Fix reported defects
Librarian or User	Perform user acceptance testing

Result

A comprehensive Test Plan was prepared for the Library Management System, including test scope, testing methodologies, test design techniques, execution strategy, entry and exit criteria, and deliverables.

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